ITI201: Fundamentals of Internet of Things

Rational

The course on the Internet of Things (IoT) is imperative for students due to its transformative potential and far-reaching societal benefits. IoT, a cutting-edge technology, offers cross-domain relevance, aiding sectors like healthcare, agriculture, and industry. It enhances operational efficiency, promotes data-driven decision-making, and addresses critical concerns like security and privacy. Furthermore, IoT knowledge fosters innovation, encouraging entrepreneurship and sustainable practices, thus positively impacting the environment. With global connectivity and collaborative possibilities, students can engage in international projects, contributing to global progress. It equips students to navigate the evolving digital landscape, innovate solutions, and significantly influence societal progress through the potential of IoT technology.

Learning outcomes

This course aims to equip students with the knowledge and skills necessary to understand, design, and implement Internet of Things (IoT) solutions across various domains, ensuring efficiency, security, and innovation in IoT applications.

Main Objective

By the end of the course-tenure the students should to:

- develop a deep understanding of IoT architectures, encompassing components, networking protocols, and communication mechanisms.
- master sensor integration, actuation, and interfacing techniques for efficient data collection and real-time responsiveness in IoT applications.
- explore a variety of IoT communication protocols (e.g., RF, ZigBee, LoRa) and standards to select and implement the most suitable protocol for specific IoT deployments.
- apply IoT principles effectively in real-life scenarios, including household, agriculture, and industrial applications.
- gain expertise in IoT security, covering data privacy, access control, and secure communication to ensure system integrity and reliability.
- acquire data analytics skills for extracting valuable insights from IoT-generated data, enabling data-driven decision-making and efficiency enhancement.

- learn optimization techniques for IoT deployments, focusing on energy efficiency, device performance, and resource utilization for sustainable solutions.
- understand the concept of edge computing in IoT and its role in real-time data processing and analytics at the network's edge.
- foster a collaborative approach to IoT integration, facilitating seamless interactions between devices to maximize IoT's potential in diverse applications.

Course content

Unit 1: Introduction

· Terminologies in IOT
· Components of IOT: Sensors, actuators, interfaces, networks, APIs, cloud
· IOT Market
· Trends and the future of IOT with real life examples.
· Data management analytics
· Data mining and manipulation
· Role of IOT in manufacturing processes
· Use of IOT in plant maintenance

Unit 2: IOT components

· Pros and cons of different architectures of IOT
· IOT System components: Sensors, gateways, routers, modem, cloud brokers, servers.
· Wireless sensor network and wired sensor network (WSNs) design
· IOT components integrations

Unit 3: Sensors and Interfacing

· Introduction to sensors and transducers
· Classification and roles of sensors in IOT
· Design and architecture of sensors
· Types and role of actuators
· Sensors with different protocols such as HART, MODBUS-Serial & Parallel.

Unit 4: Protocols and cloud

· Different IOT connection protocols: RF, Wi-Fi. Wi-Fi direct, ZigBee, Bluetooth
· Communication Channels: GSM/GPRS, 2G, 3G, LTE, WiFi.
· IOT protocols and architecture: messaging protocols for IOT (MQTTS)

· Comparison of various Low-power WAN (LPWAN) protocols like Sigfox, LoRA and LoRAWAN, Weightless, NB-IoT, LTE-M.

Unit 5: Industry 4.0 Architecture

· Object Linking and Embedding (OLE) for Process Control (OlePC).
· OlePC and distributed component object model (DCOM) diagnostics
· OlePC Unified Architecture (OlePC UA) and Security.
· Introduction to web security.
· Conventional web technology and relationship with IOT.
· Vulnerabilities of IOT.
· Security requirements: Identity establishment, access control, message integrity and availability
· Network security techniques

Unit 6: IoT Analytics and Applications

· Role of Analytics in IOT
· Data visualization techniques: Statistical methods, Visualization with python
- Applications: E-Health Body Area Networks, City Automation, Automotive Applications, Home Automation, Plant Automation, Manufacturing process automation

Teaching and Learning Methods

Lectures:45Hrs Tutorials:15Hrs, Labs:30Hrs, Field: 15Hrs, Assessments:15Hrs, NATIONAL HRS: 120

E1.9.4 Assessment

- Continuous Assessment 40%
  - 20% for hands-on assessments
  - 20% for non-hands-on assessments
- Final Exams 60%
  - 20% for hands-on assessments
  - 30% for theory assessments

E1.9.5 Recommended Readings

- Minoli, D. (2013). Building the Internet of Things with IPv6 and MIPv6: The evolving world of M2M communications. Wiley. ISBN: 978-1-118-47347-4.
- Scholz-Reiter, B., & Michahelles, F. (Eds.). (2011). Architecting the Internet of Things. Springer. ISBN: 978-3-642-19156-5.

- Chaouchi, H. (2010). The Internet of Things: Connecting objects to the web. Wiley. ISBN: 978-1-84821-140-7.
- Hersent, O., Boswarthick, D., & Elloumi, O. (2012). The Internet of Things: Key applications and protocols (2nd ed.). Wiley. ISBN: 978-1-119-99435-0.
- Deloitte University Press. (2015). Inside the Internet of Things (IoT). Deloitte.
- Vermesan, O., & Friess, P. (Eds.). (2014). Internet of Things: From research and innovation to market deployment. River Publishers. ISBN: 978-87-92329-73-5.
- Wainwright, P., & Ashton, K. (2015). Five thoughts from the father of the Internet of Things. Deloitte Insights.
- RedLion. (2017). How protocol conversion addresses IIoT challenges (White Paper).